

SENSOR CONFIGURATION FILE

LKA | AEB | Endurance

Team Equinox

aBAJA SAEINDIA 2026 — Autonomous Vehicle System

1. Overview

This document provides the complete sensor configuration for all three dynamic events — Lane Keep Assist (LKA), Autonomous Emergency Braking (AEB), and Endurance — covering camera exposure settings, radar interface parameters, CarMaker DVA quantities, vehicle model parameters, and embedded compute specifications. All sensor inputs are raw or minimally processed data. No ground-truth lane state is used as an algorithm input (Rule R1 compliant).

Sensor	Event	Interface	Primary Output
Camera RSI	LKA / Endurance	TCP — localhost:2210	640×480 RGB frames at ~30 Hz
Automotive Radar RSI	AEB / Endurance	TCP/UDP — localhost:2211	Range (m), velocity (m/s), bearing
IMU	Endurance	ROS 2 topic	Orientation, angular velocity, linear acceleration
Wheel Speed Sensors	Endurance	Hall Effect → MCU	Per-wheel rotational speed
Steering Encoder	Endurance	Absolute encoder → MCU	Real-time steering angle feedback
Wheel Encoder	Endurance	Rotary encoder → MCU	Odometry and longitudinal velocity
EKF Fusion Layer	Endurance	Software (100 Hz)	Fused vehicle state estimate

Camera RSI Sensor — LKA & Endurance

2.1 Camera Specifications

Parameter	Value	Description
Sensor Type	RGB Camera (Camera RSI)	Primary LKA perception sensor
Resolution	640 × 480 px	Optimised for real-time embedded processing at 30 Hz
Frame Rate	~30 Hz	Synchronised with the unified 30 Hz control loop
TCP Host / Port	localhost : 2210	CarMaker RSI stream endpoint
Output Format	RGB over TCP	Raw image data — no pre-processing applied by CarMaker
Mount Position	Front — Forward Facing	Lane perception and forward navigation
Processing Stack	OpenCV + ROS 2 Humble	Computer vision pipeline on NVIDIA Jetson Orin Nano
Primary Function	Lane Detection (LKA)	Provides lateral error and steering correction signal

2.2 Camera Exposure Settings

Default CarMaker camera parameters produced severely underexposed images (pixel values 10–30 out of 255, approximately 5 stops below adequate exposure). This made it impossible for any classical CV algorithm to distinguish lane markings. The following corrections were applied and validated:

Parameter	Default	Corrected	Rationale
ISO	100	800	+3 stops of sensor sensitivity. Substantially increases brightness without motion artefacts at 30 km/h.
F-Stop	11.0	2.8	+4 stops via wider aperture. Combined with ISO correction, provides adequate exposure for CV algorithms.
Shutter Speed	250 Hz	250 Hz	Unchanged — motion blur is negligible at 30 km/h on the test track.
Tone Mapping	None	Filmic	Compresses HDR to LDR dynamic range. Without this, highlights clip to white even with corrected ISO/aperture.

The near-black rendering was caused entirely by the default exposure and tone-mapping chain — not by scene lighting or sun-angle parameters. Correcting ISO, F-Stop, and tone mapping fully resolved the issue without modifying any environment settings.

2.3 LKA Perception Pipeline

The camera feed drives an 8-stage classical computer vision pipeline executed each control tick:

#	Pipeline Stage	Output
1	Image acquisition from Camera RSI TCP stream	Raw 640×480 BGR frame
2	Grayscale conversion	Single-channel intensity image
3	Binary thresholding	Binary lane mask
4	Bird's-Eye View (BEV) perspective warp — trapezoid → rectangle	Top-down lane image
5	Bottom-half histogram → sliding window search (9 windows, ±50 px)	Lane pixel coordinates
6	1st-order polynomial fit to left and right lane pixel sets	Lane polynomials
7	Lane centre estimation → lateral error (metric, via px-per-metre calibration)	Lateral error (m)
8	Lookahead point at 5.0 m ahead on fitted curve → Pure Pursuit + PID steering	Steering cmd (rad)

Automotive Radar RSI Sensor — AEB & Endurance

3.1 Radar Specifications

Parameter	Value	Description
Sensor Type	Automotive Radar (Radar RSI)	Primary AEB obstacle detection sensor
Detection Output	Distance + Relative Velocity	Range (m) and closing speed (m/s) per detected obstacle
TCP Host / Port	localhost : 2211	CarMaker RSI radar stream endpoint
Loop Rate	30 Hz	Matched to unified control loop rate
Mount Position	Front — Forward Facing	Obstacle detection in vehicle's path
Control Algorithm	Fuzzy Logic (Mamdani)	Maps TTC + closing speed to braking intensity
Primary Function	AEB — Obstacle Detection	Generates braking command for emergency intervention
Operational Benefit	Low-visibility robust	Functions in rain, dust, and low-light conditions

3.2 AEB Processing Pipeline

Radar data is processed each control tick through a 4-stage pipeline before a braking command is issued:

#	Stage	Processing	Output
1	Packet reception	radar_client receives raw RSI packet from TCP stream	Raw range, velocity, bearing
2	Obstacle extraction	radar_processor parses packet; validates data; extracts range and closing speed	Range (m), speed (m/s)
3	TTC computation	TTC = range ÷ closing speed; obstacle classified: CLEAR / CAUTION / CRITICAL	TTC (s), threat class
4	Fuzzy braking	aeb_controller fuzzifies TTC + speed; applies rule base; defuzzifies to intensity	Braking intensity (0–100%)

3.3 Fuzzy Logic Controller — Membership Functions

Input / Output	Membership Functions	Description
TTC (input)	FAR (>3 s) / MID (1.5–3 s) / NEAR (<1.5 s)	Temporal proximity classification
Closing Speed (input)	SLOW (<2 m/s) / MED (2–5 m/s) / FAST (>5 m/s)	Rate-of-approach classification
Braking Intensity (out)	NONE (0%) / PARTIAL (20–60%) / FULL (60–100%)	Defuzzified braking command

3.4 AEB Threat Zones

Zone	TTC	Closing Speed	Response
CLEAR	> 3.0 s	Any	No braking. LKA operates normally.
CAUTION	1.5 – 3.0 s	> 2.0 m/s	Partial braking (20–60%). LKA continues concurrently.
CRITICAL	< 1.5 s	Any	Full braking (60–100%). LKA steering suspended.

Vehicle State Sensors — Endurance

4.1 Sensor Suite

The Endurance event requires sustained autonomous operation. Five vehicle state sensors, fused by an Extended Kalman Filter (EKF) at 100 Hz, provide the localisation and state estimation necessary for reliable long-run performance.

Sensor	Type / Technology	Interface	Purpose
IMU	Inertial Measurement Unit	ROS 2 topic	Orientation, angular velocity, linear acceleration
Wheel Speed Sensor	Hall Effect Sensor	Motor controller MCU	Per-wheel rotational speed for slip and velocity
Steering Encoder	Absolute Encoder	MCU → ROS 2	Real-time steering angle feedback for closed-loop control
Wheel Encoder	Rotary Encoder	MCU → ROS 2	Odometry and longitudinal velocity estimation
Motor Controller Sensor	Hall Effect Sensor	Integrated in MCU	Motor rotational feedback for speed control
EKF Fusion Layer	Software — 100 Hz	ROS 2 node	Fuses all above signals for state estimate and localisation

4.2 DVA Quantities Read by Algorithm

The following CarMaker DVA quantities are read at each control tick. These are standard simulator state outputs — they do NOT include any lane-position or road-deviation ground truth (Rule R1 compliant).

DVA Quantity	Unit	Event Used	Use in Algorithm
Vhcl.v	m/s	LKA / AEB / Endurance	Vehicle speed — scales Pure Pursuit lookahead distance and PID output gain
Car.YawRate	rad/s	Endurance	Yaw rate — logged to CSV for post-run stability analysis
Car.ay	m/s²	Endurance	Lateral acceleration — logged to CSV; not used in control loop

Road Sensor DVA quantities (Car.Road.Route.DevDist, Sensor.Road.Vhcl.VehSensor_1.Route.Deviation.Dist) are present in the vehicle model but are NOT read by the algorithm — reserved for post-run validation benchmarking only.

5. Embedded Compute Platform

The full autonomous stack — LKA, AEB, EKF fusion, and endurance state machine — is deployed on an NVIDIA Jetson Orin Nano 8GB. The platform runs ROS 2 Humble for distributed inter-node communication and is designed for real-world deployability beyond the simulation environment.

Parameter	Value	Role / Notes
Compute Platform	NVIDIA Jetson Orin Nano 8GB	Embedded autonomous AI compute module
Operating System	Ubuntu Server (Canonical)	Base OS for ROS 2 deployment
Middleware	ROS 2 Humble	Distributed inter-node robotics communication
Power Supply	19 V Regulated	Vehicle-compatible power input
Simulation Platforms	IPG CarMaker, CARLA, RViz	Validation, visualisation, and testing
Control Frequency	30 Hz	Unified LKA + AEB real-time loop rate
EKF Frequency	100 Hz	Sensor fusion update rate — independent of control loop
Camera Perception	~30 FPS	Matched to Camera RSI TCP stream rate

6. Vehicle Model Parameters

All vehicle parameters used in the control algorithms are taken directly from the CarMaker Demo_IPG_CompanyCar physical model specification. No estimated or assumed values are used in any pipeline.

Parameter	Value	Source / Use
Vehicle Model	Demo_IPG_CompanyCar	CarMaker Data/Vehicle/Examples/
Mass	1,391 kg	Vehicle file specification — inertia reference
Wheelbase	2.622 m	Used in Pure Pursuit steering angle computation (LKA + Endurance)
Track Width	~1.54 m	Vehicle file specification — lateral stability reference
Steering Ratio	16 : 1	Maps road-wheel angle to steering wheel — used in Pure Pursuit
Target Speed	30 km/h (8.33 m/s)	Set in equinox_config.py; enforced by CarMaker throttle APO channel

Max Wheel Angle	±0.20 rad	Physical limit enforced in software — applied in all LKA + Endurance modes
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7. Compliance with aBAJA SAEINDIA 2026 Rules

The Team Equinox sensor configuration and autonomous architecture comply with all event rules for LKA, AEB, and Endurance. Compliance is verified against each applicable rule below.

Rule	Requirement	Status
R1	No ground-truth object states as algorithm inputs	✓ COMPLIANT — Camera RSI and Radar RSI raw outputs only; DVA lane quantities not read
R2	No simulator-internal APIs exposing privileged world state	✓ COMPLIANT — APO interface used for control outputs only; no scene graph access
R3	Algorithm inputs from configured sensor models only	✓ COMPLIANT — All perception from Camera RSI and Radar RSI TCP streams
R4	Approved sensor types only (Camera, LiDAR, RADAR, IMU, GNSS)	✓ COMPLIANT — Camera RSI and automotive radar are both approved sensor categories
R5	Sensor config validated by Technical Review before presentation	✗ PENDING — Submit this document to Technical Committee
B.1	Real-world reproducibility — no simulation-only shortcuts	✓ COMPLIANT — BEV + Sliding Window and fuzzy logic AEB deployable on embedded hardware
B.5	Min real-time rates: Perception ≥ 10 Hz, Control ≥ 20 Hz	✓ COMPLIANT — Full LKA + AEB pipeline runs at 30 Hz; EKF at 100 Hz